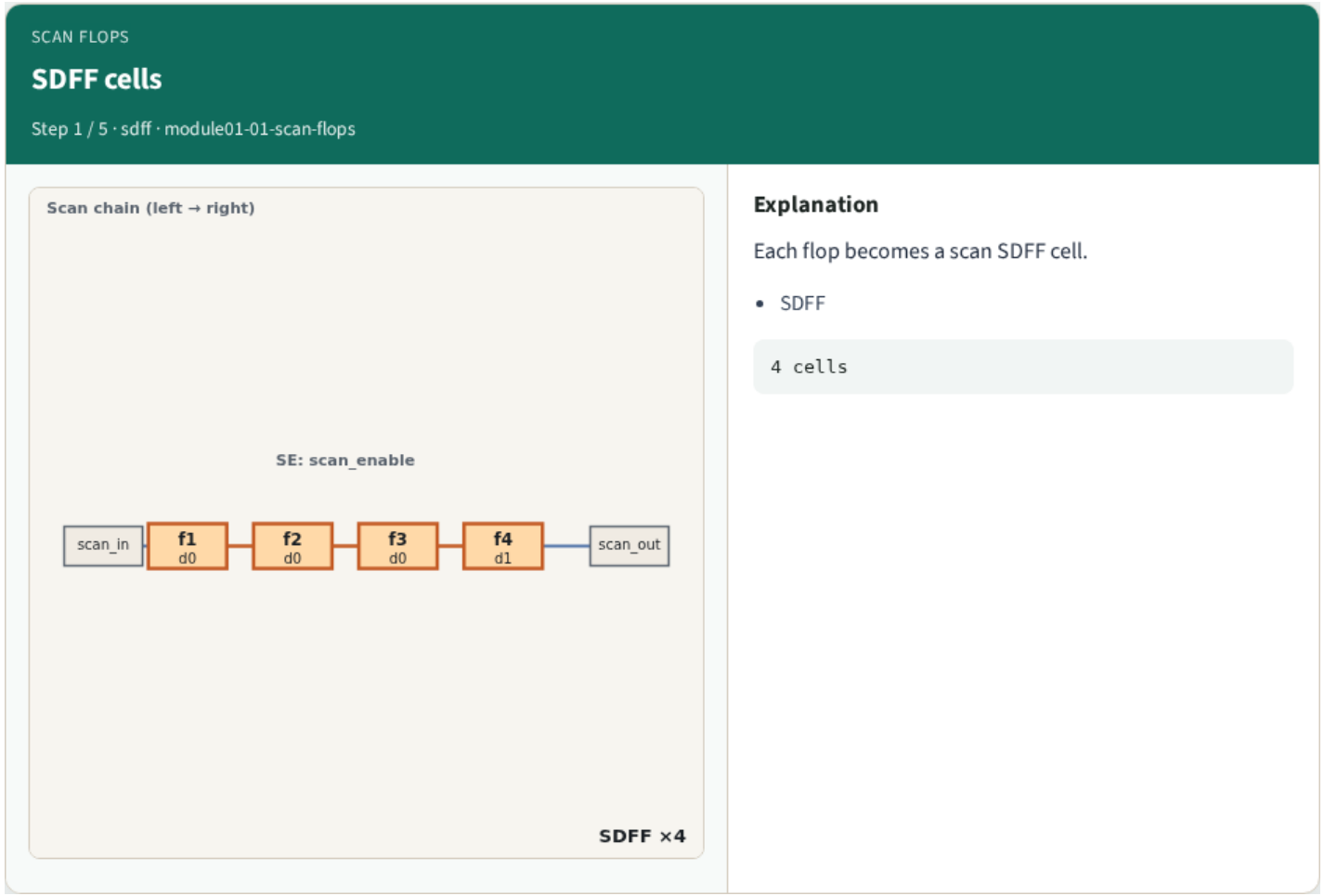
Why scan flops

Every flop becomes a scan SDFF in this full-scan toy

The idea

- GOLDEN: scan_cell_count=4

SDFF cells



SDFF cells

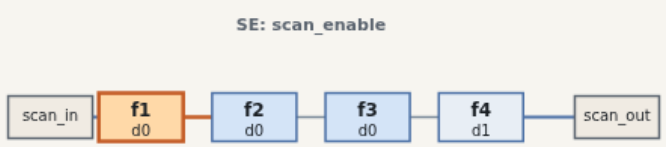
Flop f1

SCAN FLOPS

Flop f1

Step 2 / 5 · f1 · module01-01-scan-flops

Scan chain (left → right)



f1 d0

Explanation

f1: domain d0, clk=clk.

- f1

```
d0
```

Flop f1

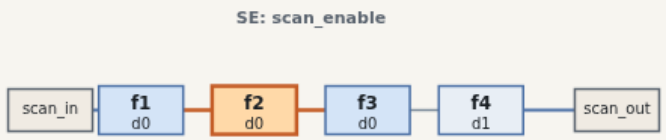
Flop f2

SCAN FLOPS

Flop f2

Step 3 / 5 · f2 · module01-01-scan-flops

Scan chain (left → right)



f2 d0

Explanation

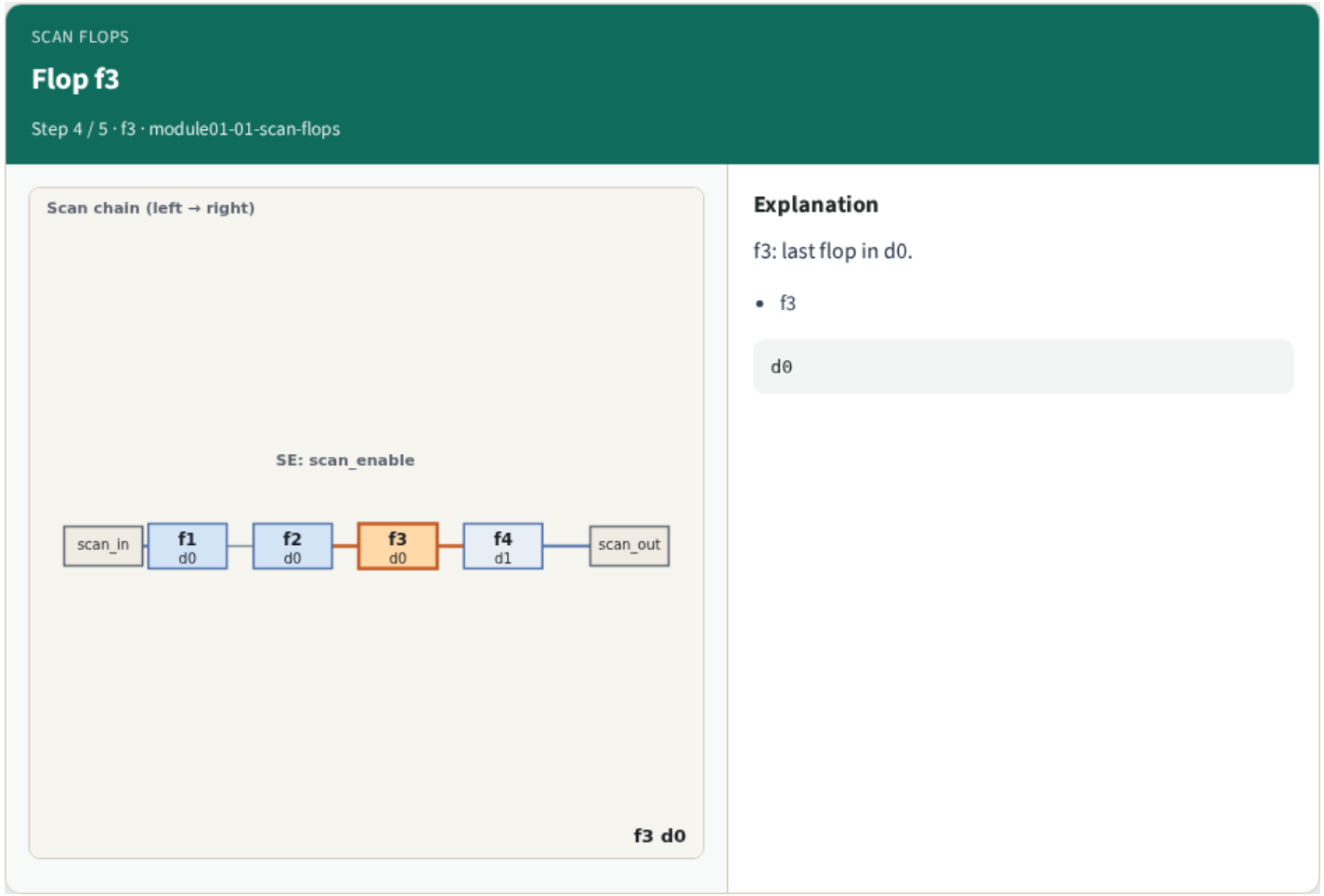
f2: Q → n2 feeds f3.

- f2

```
d0
```

Flop f2

Flop f3



Flop f3

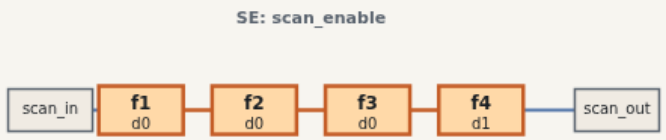
Golden

SCAN FLOPS

Golden

Step 5 / 5 · golden · module01-01-scan-flops

Scan chain (left → right)



Explanation

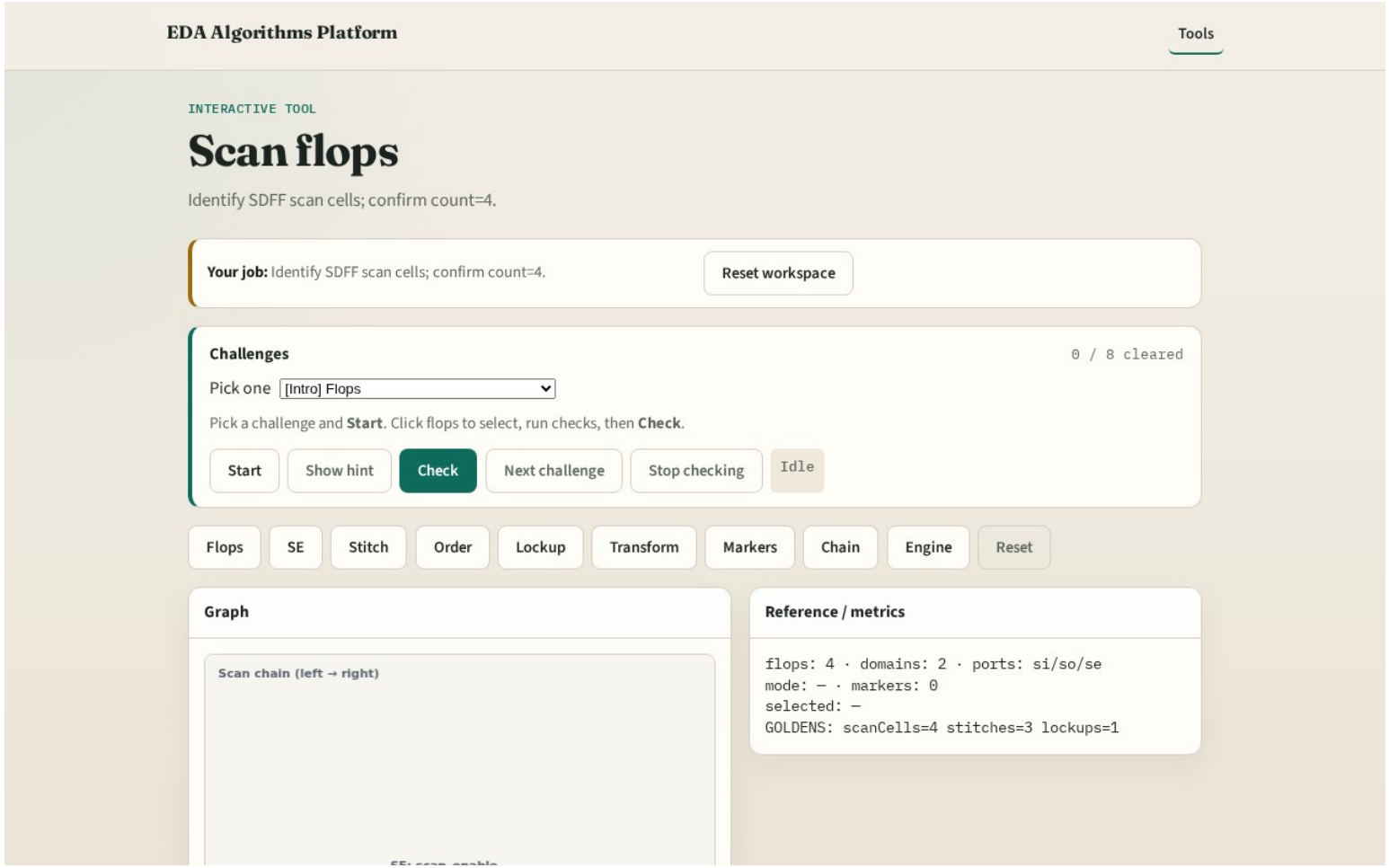
GOLDEN: scan_cell_count=4.

- GOLDEN

```
4
```

Golden

Browser lab track



Browser lab starter

Implement track

- Call identify_scan_flops(); assert count==4

Pitfalls

- Counting only functional flops
- Missing SE hook

Your turn

- Next: scan enable ports